AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

COURSE PLAN — SEMESTER: Spring 2025-2026

I. Course Core and Title

CSC 4180: Introduction to Data Science

II. Credit

3 credit hours (3 hours of theory and 3 hours Lab per week)

III. Nature

Elective Course for CSE

IV. Prerequisite

CSC4121 Artificial Intelligence & Expert System
CSC2107 Introduction to Database

V. Vision:
Our vision is to be the preeminent Department of Computer Science through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context.

VI. Mission:
The mission of the Department of Computer Science of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.

VII - Course Description:

- Comprehend introduction to the modern study of data science.
- Introduce R programming language for data science.
- Discuss various process for data collection and data preprocessing.
- Discuss the basic data management for data science.
- Describe basic statistics for data science.
- Discuss data visualization and basic graphs.
- Create graphs/plots using R for data visualization.
- Discuss web scrapping using R for data collection from websites.
- Implement interactive dashboard using R for data analysis.

VIII – Course outcomes (CO) Matrix:

By the end of this course, students should be able to:

COs *	Description	Domain Level ***			PO Assessed ****
		C	P	A	
CO1 **	***Analyze* the data to draw a valid conclusion following the data science principles and considering the limitations.**	4			PO-d-2
CO2	*Break down* the processing of collected data according to the data science principles.	4			PO-d-2
CO3	*Evaluate* the solution of data science problem by considering all the requirements.	5			PO-d-3
CO4	*Justify* the solution of data science problem to provide a valid conclusion.	5			PO-d-3

C: Cognitive; P: Psychomotor; A: Affective Domain
* *CO assessment method and rubric of COs assessment is provided in later section*
** *COs will be mapped with the Program Outcome (POs)*
*** *The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.*
**** *The numbers under 'PO Assessed' column represent the POs each CO corresponds to.*

IX – Topics to be covered in the class:*

Time Frame	CO Mapped	Topics	Teaching Activities	Assessment Strategy(s)
Week 1	CO1, CO2	**Introduction to Data Science** Theory • Introduction to the course • What is Data Science? • The Data Science Lifecycle • Datafication and applications • Key Roles Lab • Introduction to R and RStudio • Basic syntax, variables, and packages. • R control flow structures and user-defined functions	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project
Week 2	CO1, CO2	**Statistical thinking and Understanding Data** Theory • Types of Data (Structured, Unstructured) • Variable Types (Nominal, Ordinal, etc.) • Populations vs. Samples • Descriptive Statistics (Mean, Median, SD). • Dataset Components: Observations, Features, Labels • Sources of Data Collection (APIs, Surveys, Web Scraping) Lab • Introducing R Data Structures: • Vector, Matrix, Array • Data Frame, Factor, List	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project
Week 3	CO1, CO2	**Exploratory Data Analysis** Theory • The purpose of EDA • Understanding distributions (Normal, Skewed) • Introduction to Data Visualization principles, using Histograms, Boxplots, Scatterplots to find insights. Lab • Importing data • Calculating basic descriptive statistics • Exploring variable types and converting data types	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project
Week 4	CO3, CO4	**Data Cleaning & Wrangling, Midterm Project Planning** Theory • Handling missing values and duplicates • Tidy Data principles, filtering, merging, and reshaping datasets. • Data Transformation: Encoding categorical data, normalization	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project

		• Feature Engineering basics • Discussion and Planning of Midterm Projects • Reviewing project topics, datasets, expected outcomes Lab • Data cleaning with na.omit(), duplicated(). • Data wrangling with the dplyr package • Creating new variables, Normalization & scaling		
Week 5	CO2, CO3	**Introduction to Machine Learning & Regression** Theory • What is Machine Learning? • Supervised vs. Unsupervised Learning • Introduction to Linear Regression Lab • Building, interpreting, and evaluating simple linear regression model	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project
Midterm exam (Week 6)				
Week 7		Midterm Project Work Session		Viva, Report
Week 8	CO1, CO4	**Introduction to Text Data Preprocessing** Theory • Basics of text data and preprocessing (tokenization, stop word removal, stemming) • Simple text mining and word frequency analysis Lab • Advanced visualization with ggplot2 • Identifying trends, correlations, and outliers visually	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project
Week 9	CO1, CO4	**Text Mining (NLP) & Final Project Planning** Theory • Basics of NLP: What is text data? • Text Preprocessing, TF-IDF • Word Frequency and Word Clouds • Discussion and Planning of Final term Projects • Reviewing project topics, datasets, expected outcomes Lab • Web scraping with rvest. • Preprocessing with tm, tidytext • Frequency analysis and visualization	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project
Week 10	CO2, CO3	**Introduction to Dimensionality Reduction, Clustering & Data Ethics** Theory • Basics of Clustering (K-Means Algorithm) • Introduction to Dimensionality Reduction (PCA). • Real-world Applications • Ethics in Data Science (Bias, Fairness, Privacy, Transparency). **Lab** • Performing PCA on real datasets, Visualizing PCA outputs • Running K-Means clustering and interpreting clusters	Lecture, Question-answer, Lab Practice	Quiz/Exam, Project
Week	CO2,	Final Project Work Session		Viva,

11	CO3			Report
Final term (Week 12)				

* *The faculty reserves the right to change, amend, add, or delete any of the contents.*

X – Mapping of PO/PLO and K, P, A of this course:

PO Indicator ID	PO Indicators Definition (As per the requirement of WKs)	Domain	K	P	A
PO-d-2	Analysis and Interpretation of collected data to provide valid conclusion acknowledging the limitations.	Cognitive Level 4 (Analyzing)	K8		
PO-d-3	Investigate solution of complex engineering problem by synthesis of information to provide valid conclusions.	Cognitive Level 5 (Evaluating)	K8	P1 P4 P5	

XI – K, P, A Definitions

Indicator	**Title**	**Description**
K8	Research Literature	Engagement with selected knowledge in the research literature of the discipline
P1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
P4	Familiarity of issues	Involve infrequently encountered issues
P5	Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	**Description**	**Mapped POs**	**Assessment Method**	**Assessment Rubric**
CO1	Analyze the data to draw a valid conclusion following the data science principles and considering the limitations.	PO-d-2	Project	Rubric for Project
CO2	Break down the processing of collected data according to the data science principles.	PO-d-2	Quiz/Exam	Rubric for Quiz/Exam
CO3	Evaluate the solution of data science problem by considering all the requirements.	PO-d-3	Project	Rubric for Project

CO4	Justify the solution of data science problem to provide a valid conclusion.	PO-d-3	Project	Rubric for Project

XIII – Evaluation and Assessment Criteria

CO1: *Analyze* the data to draw a valid conclusion following the data science principles and considering the limitations.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Problem Analysis	Clearly demonstrate the project idea and why this problem is important to consider. Design and justify the step-by-step solution of the problem to achieve a valid conclusion from the data.				
Interpretation	Provided an analysis of the data and clearly state its outcome. Justify the outcome of the project acknowledging the limitations.				

CO2: *Break down* the processing of collected data according to the data science principles.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Content knowledge	Demonstrates the knowledge of the data science practice and principles.				
Argumentation	Articulates a position or argument for the choosing the correct practice and principles of data science.				

CO3: *Evaluate* the solution of data science problem by considering all the requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Requirements Analysis	Clearly articulates all the requirements to address the project. Presents sufficient explanation how do you accommodate all these requirements in solution design.				
Solution Design	Design a step-by-step solution of the project. Clearly demonstrate each step of the solution design considering all the requirements of the project.				

CO4: *Justify* the solution of data science problem to provide a valid conclusion.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Justification	Justify how your proposed solution is sufficient to achieve a valid conclusion from the data.				
Presentation	Clearly present your conclusion you have drawn from the data by following your proposed solution. Explain the actual outcome with the expected outcome. Draw your concluding remarks about the limitations and possible improvements.				

XIV- Course Requirements

- Students are expected to attend at least 80% class.

- Students are expected to participate actively in the class.
- For both terms, there will be at least 2 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.
- Submit report based on the given course related problems.
- Submission of assignment and/or projects should be in due time.

XV – Evaluation & Grading System*

The following grading system will be strictly followed in this class

Midterm Exam:
Term Project: 50%
Quizzes: 40%
Attendance & Performance: 10%

Final Term Exam:
Term Project: 50%
Quiz/ Lab Performance: 40%
Attendance & Performance: 10%

Semester grade: 40% midterm + 60% final term

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60
F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

** The evaluation system will be strictly followed as par the AIUB grading policy.*

XVI – Textbook/ References

- An Introduction to Data Science by Jeffrey S. Saltz and Jeffrey M. Stanton.
- R in Action by Rob Kabacoff.
- Introduction To Data Science: A python Approach to Concepts, Techniques and Applications by Laura Igual and Santi Segui.
- Lecture materials will be provided online at the course website weekly.

XVII - List of Faculties Teaching the Course

FACULTY NAME	SIGNATURE
TOHEDUL ISLAM	
Dr. ASHRAF UDDIN	
ABDUS SALAM	
KAMRUN NAHER KOLI	
RICHARD PHILIP SUPTA	

XVIII – Verification:

<table>
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Department of Computer Science

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Dr Md Abdullah Al Jubair
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Faculty of Science & Information Technology

Date:...</td><td>Approved by:

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Dr. Dip Nandi
Associate Dean,
Faculty of Science & Information Technology

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